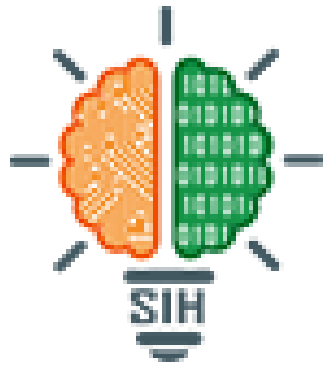
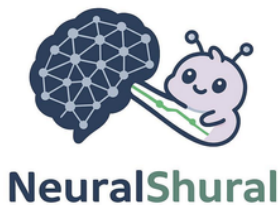




SMART INDIA HACKATHON 2025



Solution Documentation and Progress Report



Team name : **NeuralShural**



Institution : Delhi Technological University
(formerly DCE)



Problem Statement ID : **SIH25162**



Problem Statement Title : Automated
Optical Inspection (AOI) based IC marking
to identify fake marking



Theme : Smart Automation



Evaluation Round-1 Feedbacks :

1. Scrape out ample amount of OEM datasheets from internet for creating a better quality OEM-DB.
2. Make the process more robust and try to capture other points for counterfeit judgement.
3. Implement **explainability** in the entire decision making process.
4. Explain the **entire flow** through a proper flowchart.



Evaluation Round-2 Feedbacks :

1. Try out the model on **different types of ICs**
2. Implement a combined Detection + Segmentation Model in place of SAM
3. Try to make the entire workflow run in offline conditions as well



Things Improved after Round-1 :

1. OEM datasheet-DB now contains 300+ OEM sheets from major companies like TI and SL etc and it is also supported by an web-scraping tool that will continuously keep updating the DB
2. Implemented **Mixture Of VLMs** (InternVL-78B variant, InternVL-38B variant, R4B(internVL2-4B variant))
3. In parallel to the MoVLMs, **two anomaly detection modules** were applied to mark an IC defective on basis of visual appearances
4. Dataset of **6500+ hand-annotated genuine IC** procured.



Evaluation Round-2 Feedbacks :

1. Improved the report generation pipeline with increased explainability, ensured with step-wise status-update.
2. Integrated the entire workflow in a fully functional software, supported by business-grade UI platform.



Value Proposition :

1. A **fully automated**, system-agnostic workflow that eliminates all manual effort.
2. A powerful **Mixture-of-VLMs** module that delivers **unmatched accuracy** and **consistency**.
3. Built-in **parallel anomaly detection** to instantly flag even the slightest defects.
4. A completely **offline solution** that guarantees privacy and zero-dependency operation.
5. No need for high-quality cameras or extra hardware, works seamlessly with what you already have.

Problem : Manual QA is slow, error-prone, and unreliable for detecting fake or mis-marked ICs, allowing counterfeits to slip into production and cause costly failures. An automated AOI-based system is needed to accurately capture and validate IC markings against OEM standards.

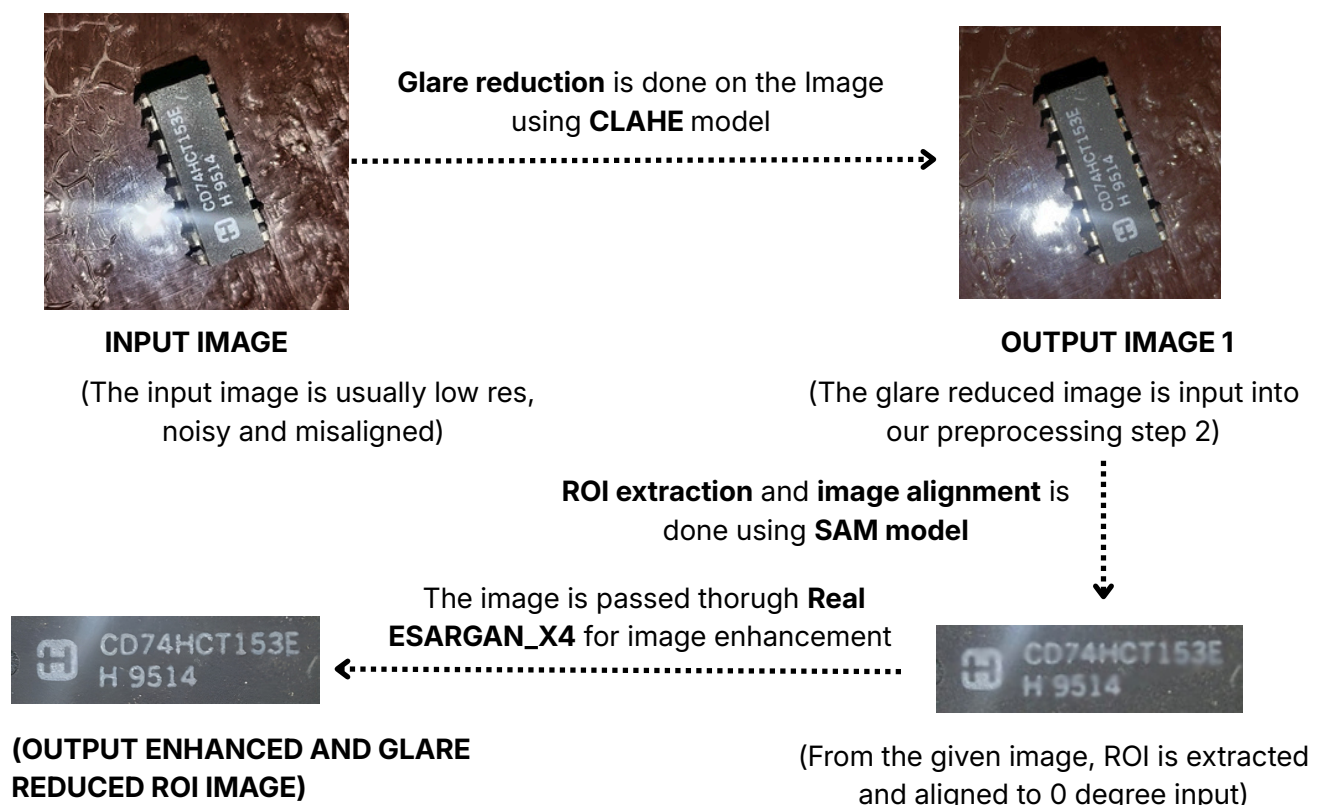
Our solution goes beyond basic marking verification by adding advanced preprocessing, specialized modules for pin counting, texture anomaly analysis, and logo detection, and an OCR+regex pipeline for serial extraction. We further integrate online datasheet retrieval, VLM-based dimension matching and explainability, and a RAG-powered knowledge base to learn from past inspection reports.



Image Preprocessing Pipeline

The preprocessing pipeline is designed to **transform raw IC images** captured from the Automated Optical Inspection (AOI) system **into high-quality, rectified inputs suitable for downstream analysis.**

The pipeline **addresses common visual challenges such as glare, low contrast, background clutter, and geometric distortion.** The complete sequence of operations is detailed below.



Text Extraction

The text extraction is done using **MSER for ROI** and **Tr-OCR for text extraction**.

The snapshot of notebbok output is attached below :



Text Region Detection Using MSER

MSER detects stable intensity regions on the IC surface that correspond to engraved or printed characters



Text Recognition Using TrOCR

The MSER-extracted regions are passed to TrOCR, which converts the visual text areas into machine-readable alphanumeric strings while handling thin engravings, font variations, and minor alignment issues.

```

=== MSER + TrOCR reads ===
[0] 1
[1] CO/4HCT153E
[2] W H9514
    
```

Final Output

The benefit of this approach is it can read even faded and worn out markings from the ICs

OEM Data Sheet Scraping

We Scrapped over 500 datasheets to make a local DB of OEM datasets for most used ICs.

The parsed JSON is converted into vector embeddings and stored back into the on-premise vector database for fast, secure future queries with full traceability.

Automated OEM Scraping (If Not Found) : If no match is found during the local OEM DB search, an automated scraper fetches the official datasheet only from verified OEM and authorized distributor sources, filtering out non-trusted pages.

OEM Parsing

All processed **OEM data sheets** are **converted into machine-readable structured JSON** formats and stored inside a local vector database.

The downloaded PDF is parsed to extract electrical specs, pin configuration, dimensions, package type, markings, and tolerances, and converted into a standardized JSON format.

We use **SOTA all-MiniLM-L6-v2** to get the **embeddings**.

Example output of the json:

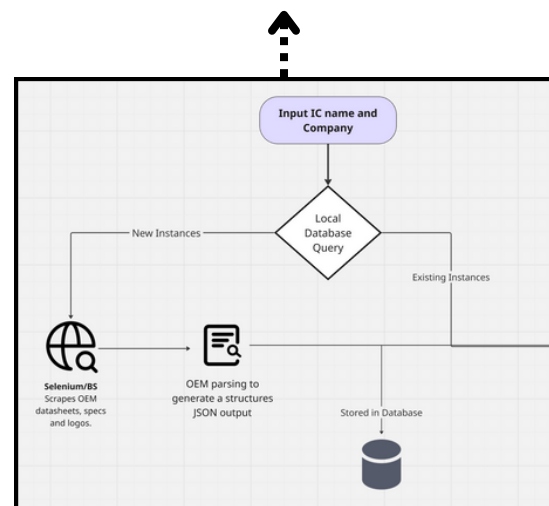
```
(myenv) PS D:\SAM_sih2025> python parser.py
{
  "extracted_text": "SN74HC04 Hex Inverter IC",
  "package": {
    "package_type": "DIP",
    "pin_count": 16,
    "length_mm": 19.69,
    "width_mm": 6.6,
    "height_mm": 8.26,
    "pin_pitch_mm": 2.54
  },
  "marking_keys": [
    "SN74HC04",
    "74HC04",
    "HC04"
  ],
  "physical_features": [
    "Notch for pin-1 identification",
    "Molded DIP package",
    "Laser or ink marking on top surface"
  ],
  "source_pdf": "datasheet.pdf",
  "extracted_at": "2025-12-08T10:05:41.904616",
  "verification_status": "OEM-Verified"
}
```

Input query ➡ **Local Database lookup**

Key Advantages

Provides a fully offline, highly secure, and scalable IC authentication system with permanent datasheet caching, fast semantic search, zero redundant fetching, and tamper-proof OEM specification access, ensuring complete IP protection and instant verification.

The OEM parsing and retrieval pipeline





The Agentic Workflow: Autonomous Verification Logic

Our system is built on a large VLM-based agent framework that operates entirely **on-device**. Within this framework, **several offline specialized agent models work independently to analyze different aspects of the input IC**. These include a **pin-segmentation and counting model**, a potential **surface-defect detection model**, and a **dimensional-mask extraction model** that verifies the physical geometry of the IC.

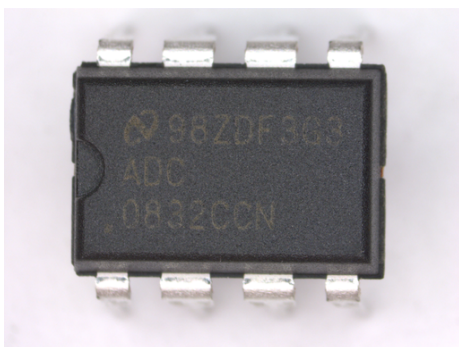
All these agent outputs are then passed to the VLM, along with the OEM-parsed datasheet information. This datasheet module has already extracted the expected pin count, package dimensions, and other critical specifications directly from the manufacturer's document, ensuring a trusted reference baseline for comparison.

By combining the multimodal agent inferences with the verified datasheet specifications, the VLM generates a final **confidence score indicating how confidently the system believes the IC is genuine**. This structured flow ensures a robust, explainable, and high-accuracy IC authenticity assessment.

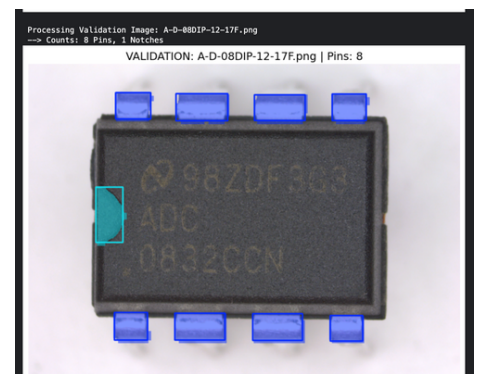


PIN SEGMENTATION AND COUNTING AGENT

We trained a custom YOLO 11s segmentation model to segment out the pins as well as potential notches in the given IC. The model was trained on dataset that we procured from Mr. Chaitanya Bhure , who is a PHD scholar doing research in this field in University of North Carolina



Input image



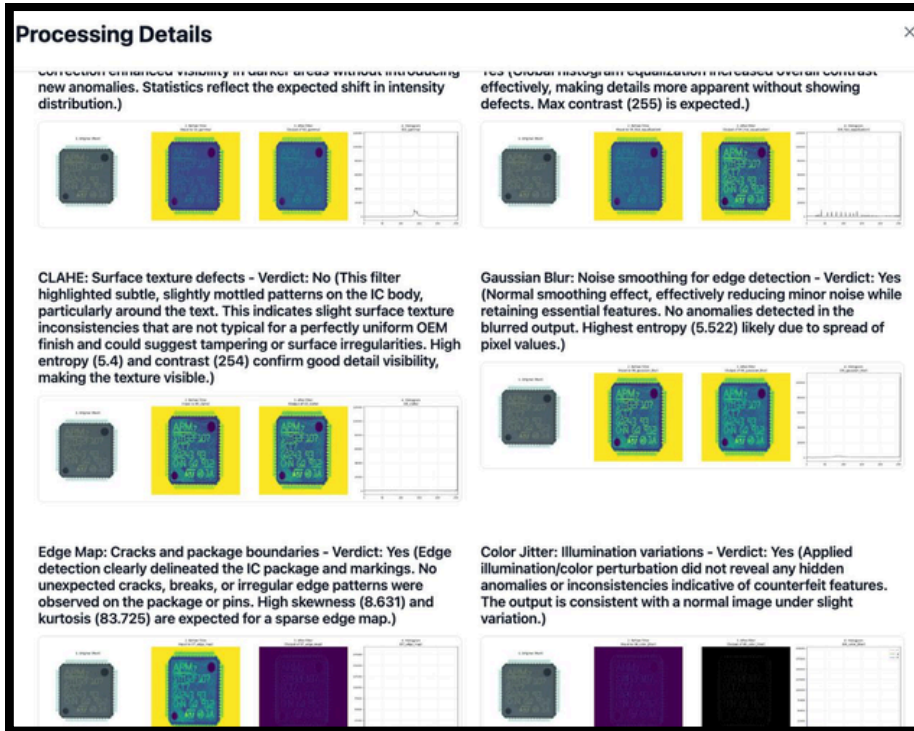
Output pins segmented out



SURFACE DEFECT DETECTION USING ALGORITHMIC APPROACH AGENT

We **conducted extensive market research on counterfeit IC detection** and identified 11 benchmark visual parameters that consistently indicate surface-quality anomalies. Using these, we apply multiple preprocessing filters-**binarization, contrast normalization, edge enhancement, and histogram-based surface profiling—to quantify texture deviations**. The resulting histogram features are analyzed to determine whether the IC surface exhibits defects such as remarking, sanding, or repackaging.

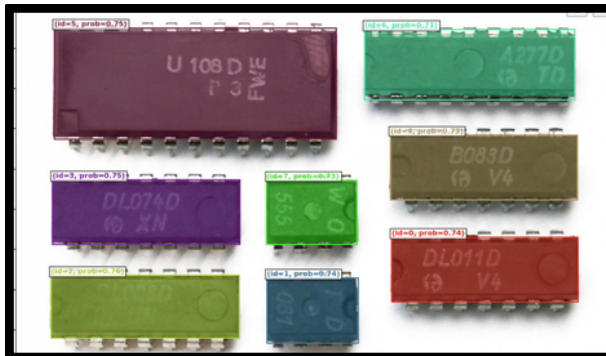
We auto-generate the final analysis of these filters for the user to trace the models predictions and understand its thinking.



An auto generated report output



IC Mask Segmentation Agent



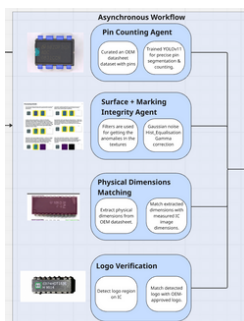
The output of the IC segmentation masks are used for dimensional analysis as many of the counterfeit ICs have dimensions different than those mentioned in the OEM sheet



LOGO DETECTION AGENT



Detected Logo to match with the OEM datasheet

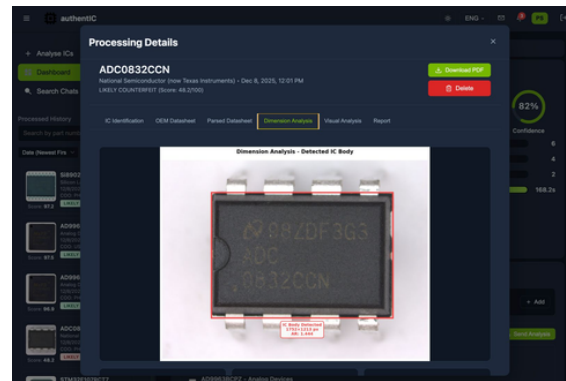
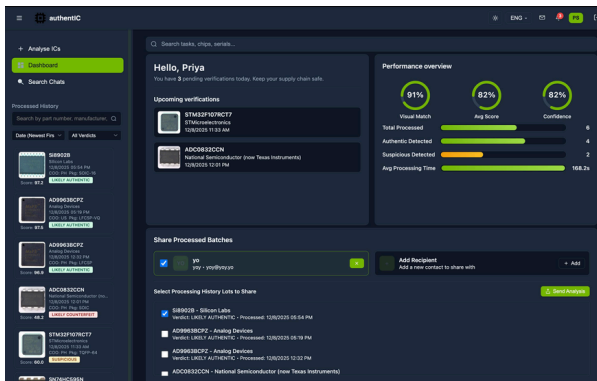


The output of these individual AI agents is combined and fed into the VLM frame work



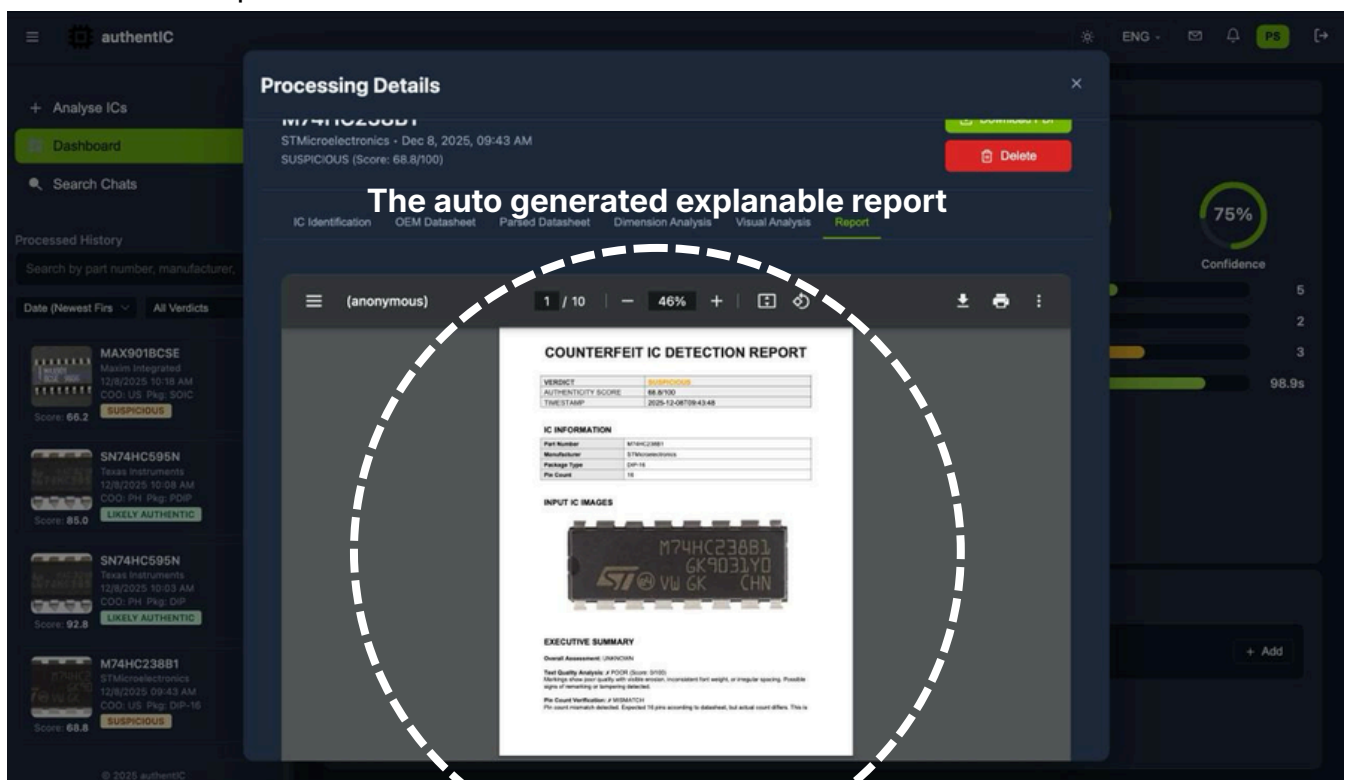
Deployment Ready Application

Our **app is fully deployment-ready**, with **both the frontend and backend live**. For VLM deployment, we secured access to the AMD MI-325X VM instance, connected via SSH, and set up the model directly on our local machine. The system is hosted through BlackRock infrastructure, with all services integrated and operational end-to-end.



Report Generation

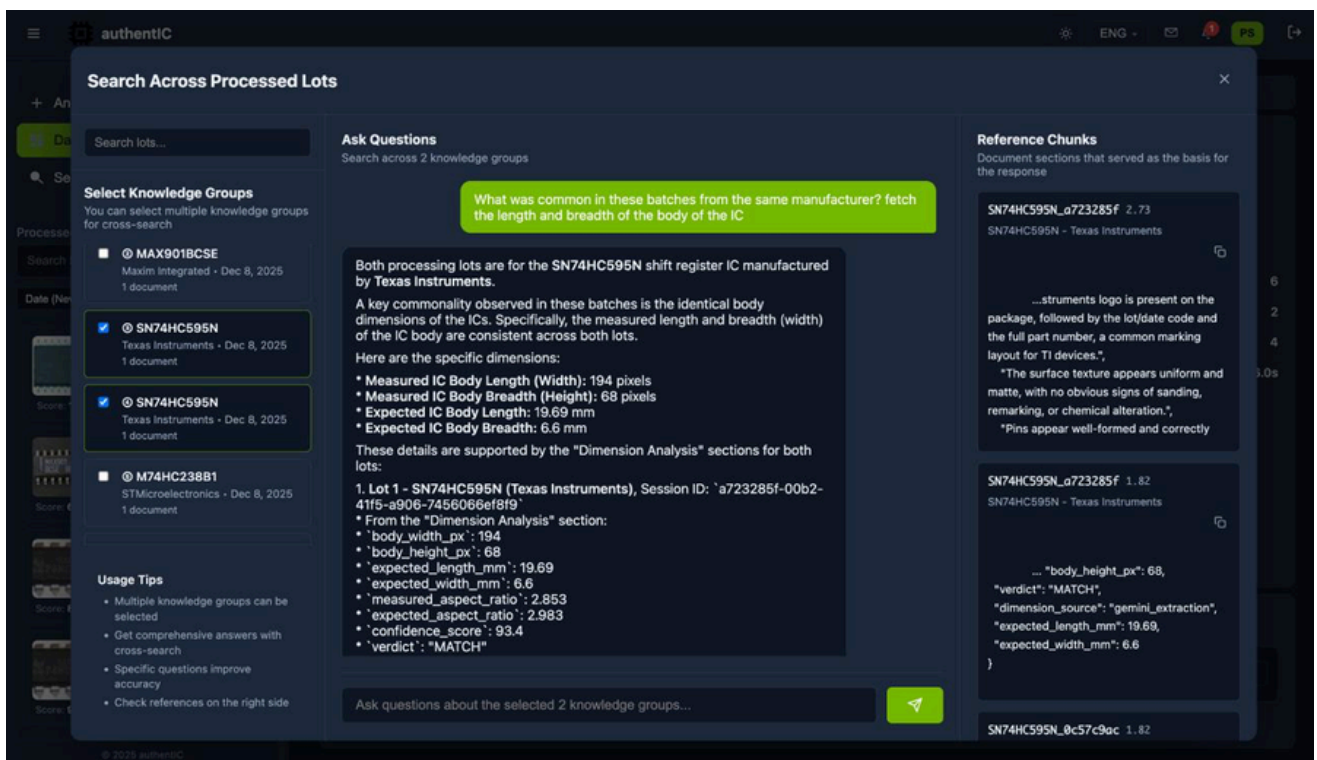
The **VLM Agentic Workflow's output is converted into a clear, explainable report** that highlights all verification steps and findings. The **report can be directly used by QA engineers, auditors, or production teams for decision-making, traceability, and compliance**. It enables fast root-cause analysis and strengthens the overall reliability of the verification process.





Intelligent Retrieval System for User Queries

The system incorporates a Retrieval-Augmented Generation (RAG) module that allows users to query past inspections and OEM documentation through natural language. When a user asks any IC-related question, the RAG engine retrieves the relevant sections from stored OEM PDFs and previous inspection records, providing context-aware answers. It highlights the exact referenced segments, enabling users to quickly verify specifications, understand marking formats, and cross-check component details with authoritative sources.



The screenshot displays the authenticIC application interface, which is a web-based tool for searching and analyzing IC components. The interface is divided into several sections:

- Search Across Processed Lots:** This section allows users to search across multiple knowledge groups. It includes a search bar and a list of selected knowledge groups:
 - MAX901BCSE:** Maxim Integrated • Dec 8, 2025 (1 document)
 - SN74HC595N:** Texas Instruments • Dec 8, 2025 (1 document)
 - SN74HC595N:** Texas Instruments • Dec 8, 2025 (1 document)
 - M74HC238B1:** STMicroelectronics • Dec 8, 2025 (1 document)
- Ask Questions:** This section provides a text input area for user queries. A sample query is shown: "What was common in these batches from the same manufacturer? fetch the length and breadth of the body of the IC". Below the input, the system provides a detailed answer:

Both processing lots are for the SN74HC595N shift register IC manufactured by Texas Instruments.

A key commonality observed in these batches is the identical body dimensions of the ICs. Specifically, the measured length and breadth (width) of the IC body are consistent across both lots.

Here are the specific dimensions:

 - * Measured IC Body Length (Width): 194 pixels
 - * Measured IC Body Breadth (Height): 68 pixels
 - * Expected IC Body Length: 19.69 mm
 - * Expected IC Body Breadth: 6.6 mm

These details are supported by the "Dimension Analysis" sections for both lots:

1. Lot 1 - SN74HC595N (Texas Instruments), Session ID: 'a723285f-00b2-41f5-a906-7456066ef8f9'

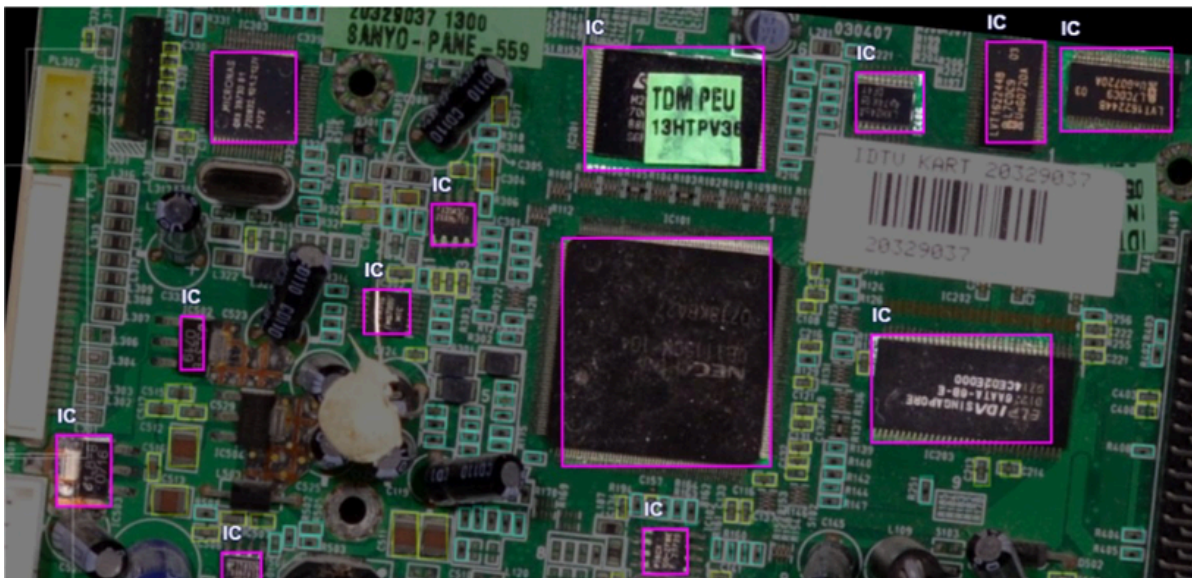
 - * From the "Dimension Analysis" section:
 - * 'body_width_px': 194
 - * 'body_height_px': 68
 - * 'expected_length_mm': 19.69
 - * 'expected_width_mm': 6.6
 - * 'measured_aspect_ratio': 2.853
 - * 'expected_aspect_ratio': 2.983
 - * 'confidence_score': 93.4
 - * 'verdict': "MATCH"
- Reference Chunks:** This section displays document sections that served as the basis for the response. It includes two chunks:
 - SN74HC595N_a723285f 2.73:** SN74HC595N - Texas Instruments. The chunk contains text about the package marking and surface texture.
 - SN74HC595N_a723285f 1.82:** SN74HC595N - Texas Instruments. The chunk contains text about the body dimensions and the "MATCH" verdict.

Additions done after mentoring and evaluation rounds

IC detection on FPGA board

Catering to the use cases highlighted by the BEL representatives during the online webinar, our system focuses on **reliably identifying ICs on FPGA boards and routing them through a dedicated verification workflow**. These components are then individually analyzed to determine their authenticity using our automated AOI-VLM pipeline. This approach ensures high-precision detection of counterfeit or mismarked ICs, addressing a critical need in high-reliability electronics. By automating the inspection process, we reduce manual dependency, improve traceability, and significantly lower the risk of defective components entering production.

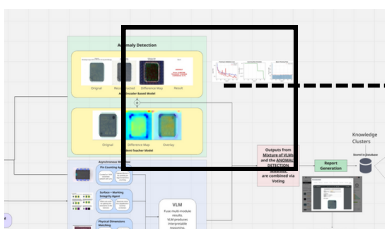
This is done so that the FPGA boards can directly be analysed for fake ICs.



Model output detected ICs

Anomaly Detection

We added a parallel anomaly-detection pipeline alongside our multi-agent VLM framework to specifically catch subtle surface defects and enable continuous self-learning from real inspection data. This uses two complementary anomaly-detection models to ensure broad coverage of visual irregularities on ICs.

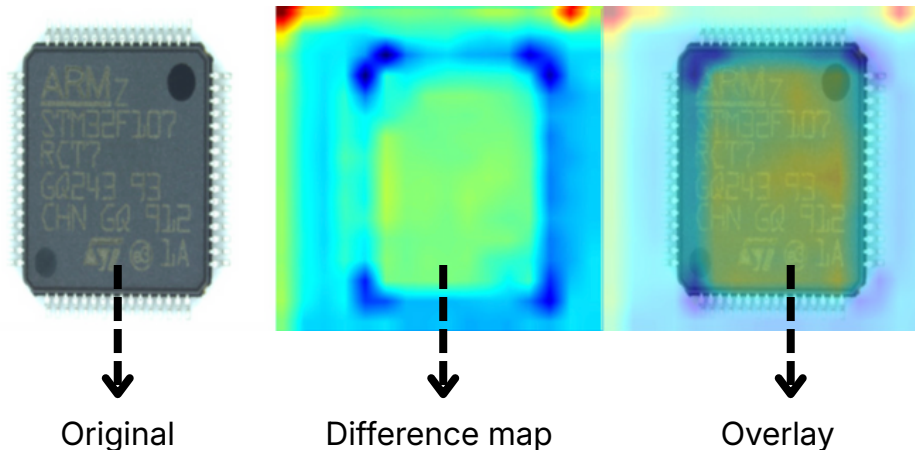


New anomaly detection pipeline added



A) Student - Teacher Based Anomaly Detection

The teacher is a general visual expert, the student is a specialist trained only on normal ICs; wherever the student fails to imitate the teacher, that region is statistically outside the normal IC distribution and is therefore marked as an anomaly.



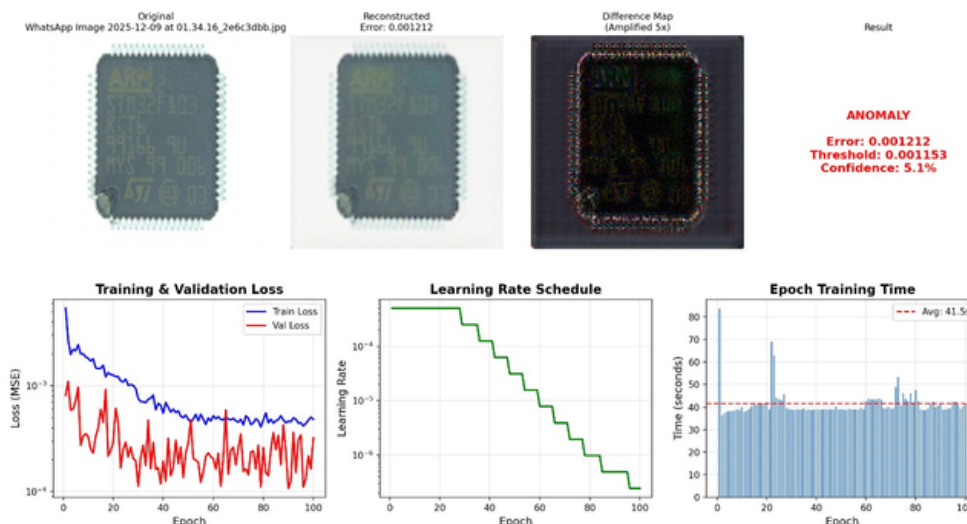
This IC does not have an anomaly



B) AutoEncoder Based Model

While researching we came across the paper : **IC-ChipNet: Deep Embedding Learning for Fine-grained Retrieval, Recognition, and Verification of Microelectronic Images** by **Md Alimoor Reza and David J. Crandall**

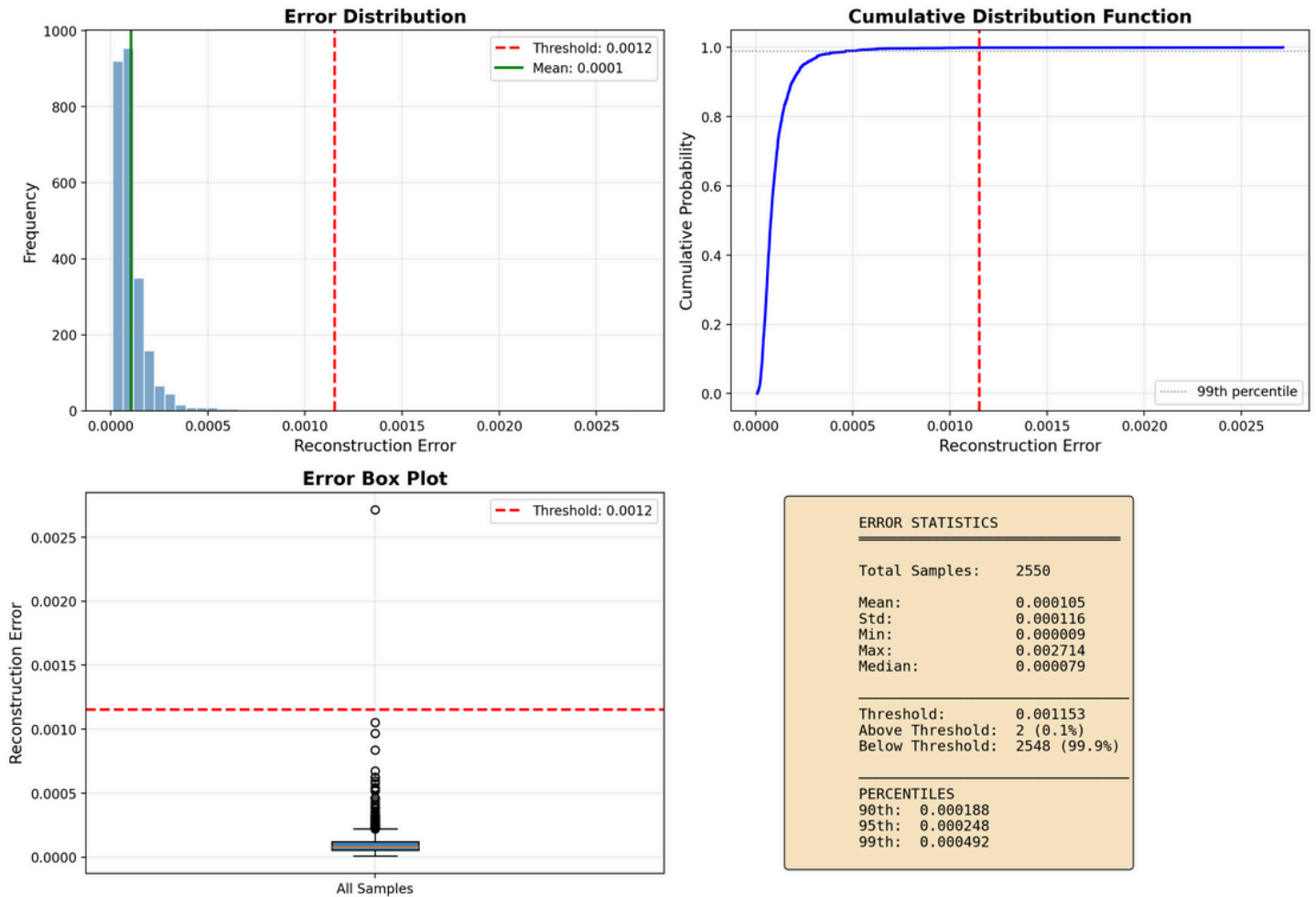
We explored multiple methods and adopted an autoencoder-based anomaly detection pipeline. We preprocess IC images through resizing, normalization, and tensor conversion, then train a five-layer encoder-decoder autoencoder using MSE loss on genuine samples. After training, we compute reconstruction errors and apply a 97.5-percentile threshold to flag unusually high-error cases as potential counterfeit





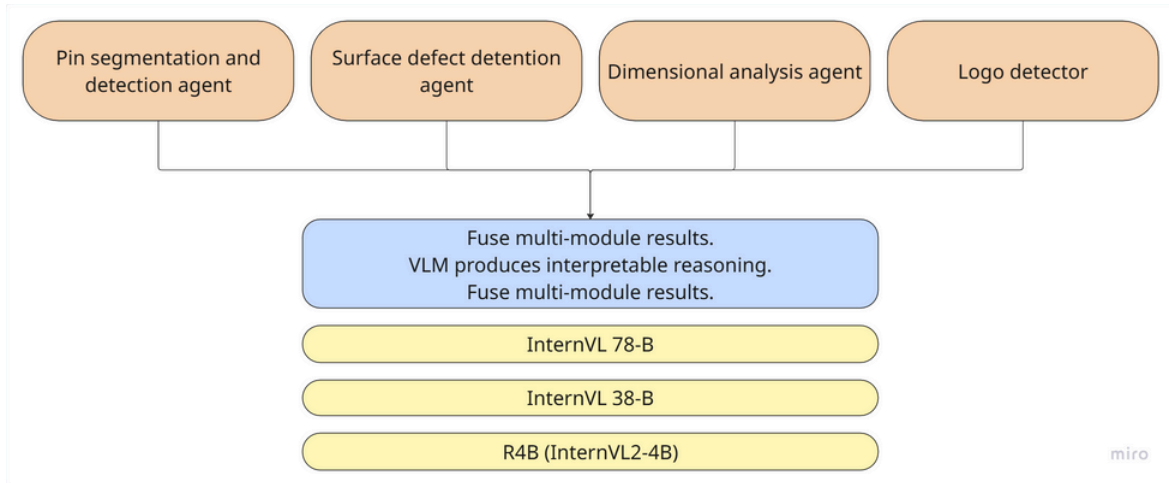
Analysis report for Autoencoder-Based Anomaly Detection

IC Anomaly Detection - Error Analysis





Mixture of VLMs



We built a Mixture-of-VLMs system combining InternVL3-78B, InternVL3-38B, and R4B. The outputs of all three models are passed to a 2-layer MLP that learns adaptive weights to decide how much each VLM should influence the final prediction.

- **R4B(InternVL2-4B) – Reliability Checker:**
Validates whether outputs generated by tools or external modules are correct.
- **InternVL3-38B – Cross-Tool Verifier:**
Verifies and compares outputs coming from multiple tools, ensuring logical and semantic consistency.
- **InternVL3-78B – Global Supervisor & Deep Reasoner:**
Acts as the high-capacity overseer, performing deep reasoning and resolving conflicts across models.

The MLP integrates these complementary roles to produce a more accurate, stable, and context-aware final output.

	InternVL2.5 78B	InternVL3 8B	InternVL3 78B	Qwen2.5-VL 72B	Other Open-Source MLLMs	Claude-3.5 Sonnet	ChatGPT- 4o-latest	Gemini-2.5 Pro
<i>Model Weights</i>	✓	✓	✓	✓	✓	✗	✗	✗
<i>Training Data</i>	✗	✓	✓	✗	-	✗	✗	✗
MMMU Multi-discipline	70.1%	65.6%	72.2% (2.1 T)	70.2%	64.5%	66.4%	72.9%	74.7%
MathVista Math	72.3%	75.2%	79.6% (7.3 T)	74.8%	70.5%	65.1%	71.6%	80.9%
A12D Diagrams	89.1%	85.2%	89.7% (0.6 T)	88.7%	88.1%	81.2%	86.3%	89.5%
ChartQA Charts	88.3%	86.6%	89.7% (1.4 T)	89.5%	88.3%	90.8%	-	-
DocVQA Documents	95.1%	92.7%	95.4% (0.3 T)	96.4%	96.5%	95.2%	-	-
InfographicVQA infographics	84.1%	76.8%	85.2% (1.1 T)	87.3%	84.7%	74.3%	-	-
HallusionBench Hallucination	57.4%	49.9%	59.1% (1.7 T)	55.2%	58.1%	55.5%	57.0%	64.1%
OCRBench OCR	854	880	906 (52T)	885	877	-	894	862
LongVideoBench Video	63.6%	58.8%	65.7% (2.1T)	60.7%	61.3%	-	-	-

Benchmarks

Hardware Requirements

Component	CPU Cores	RAM (GB)	VRAM (GB)	Storage (GB)	GPU Load	Notes
SAM 2.1	4+	4	2	2	Medium	0.5s GPU / 3s CPU per
YOLO Pin Counter	2	2	1	0.5	Low	0.1s GPU per
YOLO PCB	2	2	1	0.5	Low	0.1s GPU per
DeepSeek OCR	4	3	2	0.3	Medium	300–700 ms per
ESRGAN (opt.)	4	4	2	0.1	High	1–2s GPU per
PostgreSQL+pgvec	4+	16	0	50+	N/A	8–16 GB RAM

VLM	CPU Cores (est.)	RAM (GB)	VRAM (GB)	Storage (GB)	GPU Requirement	GPU Load
R4B VLM	8–12	32–48	24–32	20–30	Runs on 1× 4090 or 1× A100	High
InternVL3-38B (med.)	12–16	60–80	30–40	50–70	1× A100 80GB or 2× 4090	Very High
InternVL3-78B (large)	16–24	100–140	60–80	100–150	2× A100 80GB or 4× 4090	Extreme

Total Cluster-Level Resource Requirements

Resource Type	Classic Stack + DB	VLMs (R4B + 38B + 78B)	Overhead/Notes	Total Requirement
RAM (GB)	31 GB	268 GB	Headroom via server sizing	≈300 GB (minimum peak)
VRAM (GB)	8 GB	152 GB	6–10 GB (future OCR, higher-res SAM, parallelism)	≈170 GB (with safety buffer)
Storage (GB)	≈154 GB (models + DB + 50-100 GB runtime)	250 GB (upper bounds)	Logs/backups/growth → provision 8–10 TB NVMe/SSD cluster-wide	≈400 GB models/DB 8–10 TB practical
GPU Requirement	Included in classic VRAM	R4B: 1×4090 InternVL3-38B: 2×4090 InternVL3-78B: 4×4090	Per your "Requires" specs	7× RTX 4090 (168 GB total VRAM)

Servers Assigned

GPU Servers

- Each GPU server:
 - 1× RTX 4090 (24 GB VRAM)
 - 16 CPU cores
 - 64–128 GB RAM
 - 1 TB NVMe

This is enough per node for its portion of a VLM plus overhead and local cache.

- Allocation:
 - Node 1: R4B + classic models (SAM, YOLOs, DeepSeek OCR, ESRGAN).
 - Nodes 2–3: InternVL3-38B.
 - Nodes 4–7: InternVL3-78B.

VRAM- $7 \times 24 = 168$ GB

RAM- $7 \times 128 = 896$ GB

Storage- 7×1 TB NVMe = 7 TB on GPU nodes.

Cost of each components:-

RTX 4090- Rs 1,49,000
16-core CPUs- Rs 21,000
128 GB RAM- Rs 22,000
1 TB NVMe- Rs 10,000

Total Cost- Rs 14,14,000



Market and Dataset Research

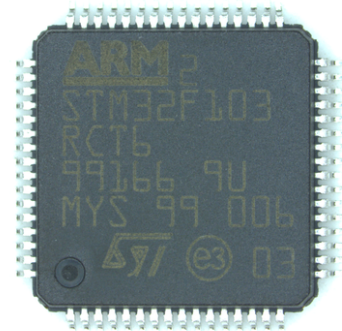
We researched for various IC datasets, in which we requested a dataset that was already labelled and being used in the research paper by Dr. Bhure, PhD University of North Carolina - It contained train, test and valid split for labelled images of counterfeit and real ICs.

This is our primary dataset that we have annotated ourselves for pin counting and segmentation

This is also the dataset which we are using as ground truth for evaluation of counterfeit and real ICs.



Name	Date Modified	Size	Kind
A-D-64QFP-08-03F.png	5 Dec 2025 at 1:10 AM	4.7 MB	PNG image
A-D-64QFP-08-04F.png	5 Dec 2025 at 1:10 AM	4.7 MB	PNG image
A-D-64QFP-08-05F.png	5 Dec 2025 at 1:10 AM	4.7 MB	PNG image
A-D-64QFP-08-06F.png	5 Dec 2025 at 1:10 AM	4.5 MB	PNG image
A-D-64QFP-08-07F.png	5 Dec 2025 at 1:10 AM	4.5 MB	PNG image
A-D-64QFP-08-08F.png	5 Dec 2025 at 1:10 AM	4.7 MB	PNG image
A-D-64QFP-08-09F.png	5 Dec 2025 at 1:10 AM	4.6 MB	PNG image
A-D-64QFP-09-01F.png	5 Dec 2025 at 1:10 AM	5.1 MB	PNG image
A-D-64QFP-09-02F.png	5 Dec 2025 at 1:10 AM	4.9 MB	PNG image
A-D-64QFP-09-03F.png	5 Dec 2025 at 1:10 AM	5.1 MB	PNG image
A-D-64QFP-09-04F.png	5 Dec 2025 at 1:10 AM	5 MB	PNG image
A-D-64QFP-09-05F.png	5 Dec 2025 at 1:10 AM	4.9 MB	PNG image
A-D-64QFP-09-06F.png	5 Dec 2025 at 1:10 AM	5 MB	PNG image
A-D-64QFP-09-07F.png	5 Dec 2025 at 1:10 AM	4.9 MB	PNG image
A-D-64QFP-09-08F.png	5 Dec 2025 at 1:10 AM	5 MB	PNG image
A-D-64QFP-09-09F.png	5 Dec 2025 at 1:10 AM	5.1 MB	PNG image
A-D-64QFP-09-10F.png	5 Dec 2025 at 1:10 AM	5.1 MB	PNG image



We also came across the IU Computer Vision Lab dataset that consists of 6500 real IC images, which we have used to train the anomaly detection model

IU Computer Vision Lab

HOME PEOPLE PROJECTS PUBLICATIONS TALKS & VIDEOS IN THE MEDIA EVENTS

IC-ChipNet: Deep Embedding Learning for Fine-grained Retrieval, Recognition, and Verification of Microelectronic Images

Md Alimoor Reza and David Crandall

Modern electronic devices consist of a wide range of integrated circuits (ICs) from various manufacturers. Ensuring that an electronic device functions correctly requires verifying that its ICs and other component parts are correct and legitimate. Towards this goal, we investigate using machine learning and computer vision to identify and verify integrated circuit packages using visual features alone. We propose a deep metric learning approach to learn a feature embedding to capture important visual features of the external packages of ICs. We explore several variations of Siamese networks for this task and learn an embedding using a joint loss function. To evaluate our approach, we collected and manually annotated a large dataset of 6,387 IC images, and tested our embedding on three challenging tasks: (1) fine-grained retrieval, (2) fine-grained IC recognition, and (3) verification. We believe this to be among the first papers targeting the novel application of fine-grained IC visual recognition and retrieval, and hope it establishes baselines to advance research in this area.

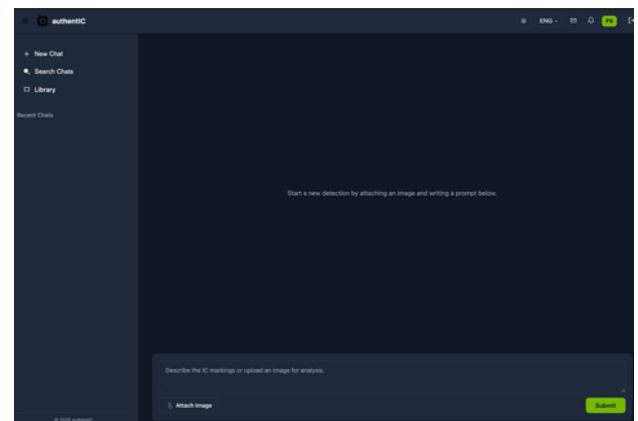
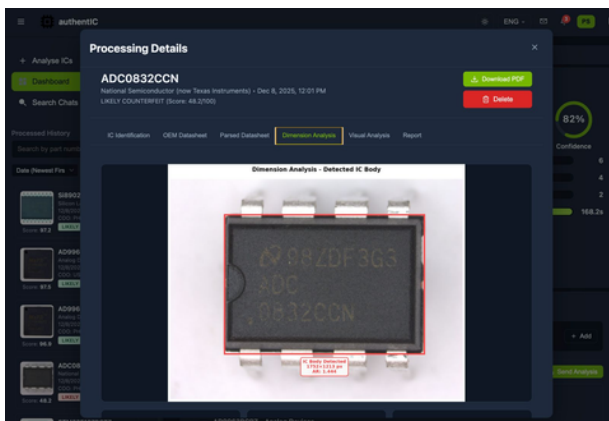




Introduction of user dashboard for consumer end counter fit IC detection

Use Case (B2B)

Electronics repair centers, component distributors, small-to-mid-scale manufacturers, and PCB assembly units face significant operational and financial risks due to counterfeit ICs entering their supply chain. Our software enables these businesses to **automate large-scale IC verification** by uploading batches of images, instantly detecting marking inconsistencies, packaging anomalies, or suspected counterfeits without relying on skilled technicians. Each inspection generates a **detailed, downloadable report** that teams can query through natural language, and the platform provides a business **dashboard showing key metrics** such as number of defects detected, types of anomalies, supplier-wise risk patterns, and inspection trends. This helps businesses improve quality control, reduce returns and failures, and maintain a more reliable component pipeline.

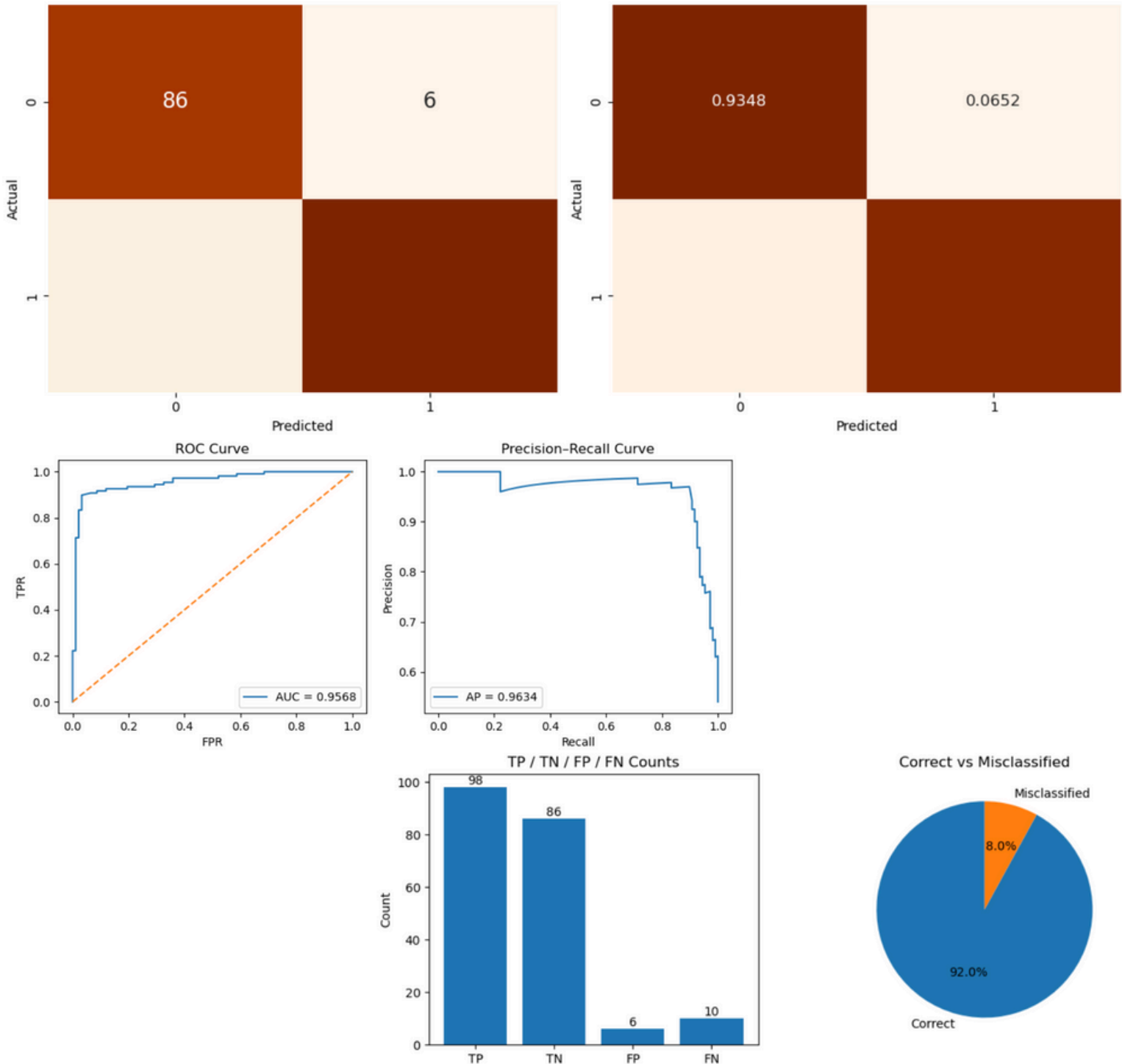


Use Case (B2C)

Counterfeit integrated circuits (ICs) are a persistent problem in India's electronics ecosystem, affecting hobbyists, small repair shops, small-scale manufacturers, and individual buyers. Currently, verifying IC authenticity requires skilled technicians, is time-consuming, and often inaccessible to non-experts. Users can also ask **natural-language questions** about the IC's functionality, pinout, or usage, making the tool a simple and accessible way for anyone to check their components before using them in projects or repairs.



Final Pipeline Results (Complete Workflow)



TP: 98 TN: 86 FP: 6 FN: 10

Accuracy: 0.92

Precision: 0.94

Recall: 0.91

F1 Score: 0.92

AUC: 0.96

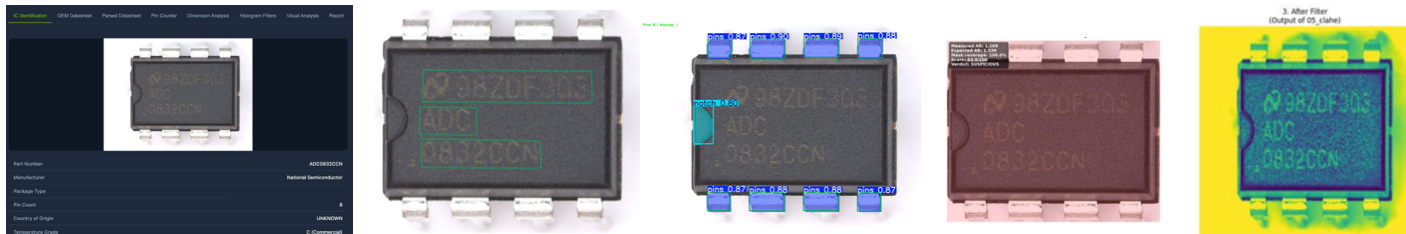
Avg Precision: 0.96

Number of predicted probabilities in (10%, 90%): 147 (out of 200)

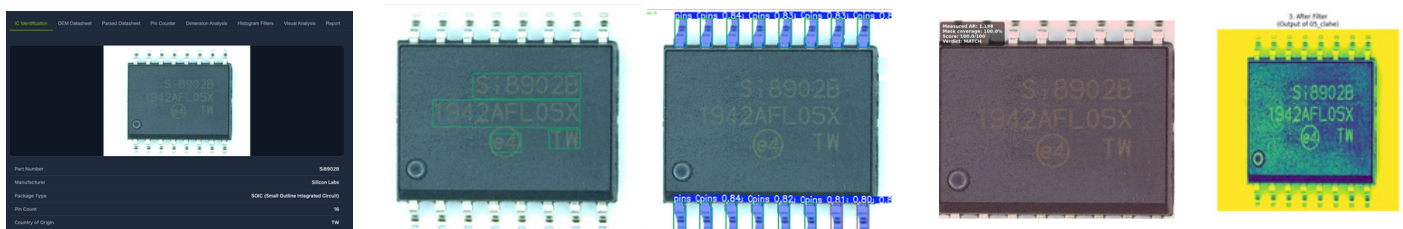


Output Flows on different IC packages

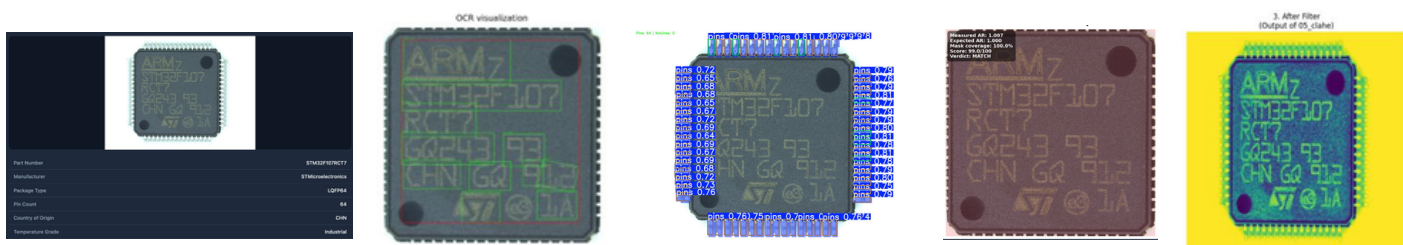
1. DIP (Dual In-line Package)



2. SOIC (Small Outline IC)



3. Quad-Lead Packages



4. Leadless Chip Packages

